Multi-observable indirect-witness chain bounding the Einstein-identity gap on a relational emergent-gravity lattice

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Abstract

The relational emergent-gravity construction's principal direct test is the per-node 4×4 Galerkin Frobenius residual, reported in the companion paper [3], which now extends to a norm-explicit bulk-percentile audit on the canonical $\mathcal{P}_5/\mathcal{P}_5N$ ten-regime ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 256, 300, 512\}$ (180 seeds canonical) supplemented by a five-regime alt-anchor cross-check $\mathcal{P}_6, \mathcal{P}_7, \mathcal{P}_8, \mathcal{P}_6 N_{128}, \mathcal{P}_8 N_{128}$ (88 seeds; total 268 seeds across 15 regimes). The witness-chain reported in the present companion note operates on the original ten-regime witness ladder $N \in \{50, 60, 64, 72, 84, 100, 200, 300\}$ (208 seeds, the witness configuration prior to the matter-localised $N \in \{128, 256, 512\}$ extensions); the eleven/fifteen-regime extended ladder of the parent paper is the principal direct test, while the present note reports witnesses on the *narrower* ladder by design, as orthogonal consistency diagnostics that supplement (do not replace) the principal direct test. Under the data-supported power-law depletion model the median, mean, p_{90} and p_{95} of the per-node relative-Frobenius residual all extrapolate to zero on the regular sector. Under the more conservative Symanzik-2 linear-in-$1/N$ audit [9] the median asymptote appears small but positive, $\text{med}_\infty = 0.019 < 0.05$ (95% CI $[0.014, 0.023]$); the parent paper [3] identifies this as a basis-truncation artefact of the Symanzik-2 basis $\{1, 1/N, 1/N^2\}$, which cannot exactly represent the inverse-spatial-dimension volumetric scaling $N^{-1/d_{\text{spatial}}} = N^{-1/3}$ of the per-percentile spectrum theorem; the basis-faithful two-power continuation $y(N) = c\, N^{-1/3} + d\, N^{-2/3}$ vanishes asymptotically and fits the same ladder at $\Delta\text{AICc} = +1.16$ relative to Symanzik-2 (statistically indistinguishable, both go to zero in the continuum). The mean, p_{90} and p_{95} asymptotes remain compatible with the $y_\infty \geq 0$ zero-floor boundary under either reading. This is the direct per-node tensor-residual audit at the bulk-percentile level. The parent paper [3] distinguishes a *transition shell* p_{98}/p_{99} (which extrapolates to small finite values under the same $N^{-1/d_{\text{spatial}}}$ basis) from a *matter-saturation core* $p_{99.5}/p_{99.9}/\sup$ (which saturates strictly to finite values, with bootstrap 95% lower-CI bounds on y_∞ positive throughout: $p_{99.5,\infty} \geq +0.044$, $p_{99.9,\infty} \geq +0.163$, $\sup_\infty \geq +0.304$). The matter-saturation core is explained by the universal density-contrast law of [3] and is structurally disjoint (Jaccard 0.02–0.03) from the source-active Ξ-row-sum top-5% support of [2].

This companion note bundles four *confirming* witnesses that bound the gap from above: (i) a five-point chirality-balance log-log cross-regime fit at $N \in \{28, 30, 36, 42, 50\}$ recovering $\alpha_{\text{gap}} = 0.6355$ within 4.7% of the cross-regime upper-bound exponent 2/3, $R^2 = 0.78$; (ii) a nine-point Ricci-scalar witness $\bar{R}(N) \to 0$ at $R^2 = 0.83$; (iii) an off-diagonal spatial-stress multi-N decay with $\alpha = 2.54$ on the extension lattice ladder; and (iv) a Look-Elsewhere Bonferroni multiplicity correction on the four-component $\Lambda_{\mu\nu}$ identification under independent regressor choices. The within-regime sup-decay exponent is identified separately at $\alpha_\xi^2 = 81/100 = 0.81$ (Section 2); the cross-regime upper-bound 2/3 and the within-regime structural value α_ξ^2 are not the same number and are bundled under distinct ladders. Each witness is bundled with its reproducibility certificate and is reported as an orthogonal consistency diagnostic supplementing the direct bulk-percentile tensor-residual audit of the parent

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paper; the chirality-balance fits remain weaker statistical witnesses ($R^2 = 0.78$ on the five-point cross-regime fit, $R^2 = 0.68$ at fixed $\alpha = 2/3$ on the within-regime fit) and are reported as supporting rather than load-bearing evidence; the load-bearing direct test is in the parent paper. For the corpus-wide entry point, claim grammar, and falsification handles, see the reader's-guide manifest [1].

1 Introduction and scope

What this note does and does not claim. This note is a topic-focused companion to the parent emergent- gravity closure paper [3]. The parent reports the per-node 4×4 Galerkin Frobenius residual as the *direct* test of the Einstein-identity gap on the canonical lattice ladder ($N \in \{18, .., 84\}$, median residual 0.026 at $N = 84$ under both blind-fit and rational-closure structural $\Lambda_{\mu\nu}$). This note bundles four *indirect* witnesses that bound the same gap from above on the cross-regime ladder at the upper-bound exponent 2/3. We do not claim that any single indirect witness is a complete proof, nor that the cross-regime 2/3 exponent is the same as the within-regime sup-decay exponent (the latter is the structural $\alpha_\xi^2 = 81/100$, see Section 2); we claim the combined chain plus the direct Galerkin test of the parent paper is the strongest available evidence at the present audit.

Honest caveats acknowledged from the parent paper. The parent paper [3] states five caveats explicitly: (i) the five-/eight-/nine-point diagnostics are multi-observable consistency rather than a direct Δ_E proof; (ii) free-fit power-law exponents are scattered across diagnostics, and the data are compatible with but do not uniquely select the cross-regime upper-bound exponent 2/3 (the within-regime sup-decay exponent is identified separately at $\alpha_\xi^2 = 81/100$ on the canonical-physics ladder, see Section 2; the two exponents are not the same number); (iii) the $N = 18$-skip variant of the Ricci-scalar fit improves R^2 but is methodologically a finite-size selection; (iv) the chirality–Δ_E bridge is an empirical correlation; the closed-form lattice Atiyah–Singer [6] theorem on the discrete relational geometry is supplied at the topological-character level by the companion atiyah-singer paper [4] via overlap-Dirac (machine-precision identity $\mathrm{ind}_{\mathrm{rel}} = Q_{\mathrm{top}}$); (v) the source-side check at $\Lambda_{\mathrm{lat}} = 0.314$ is extracted as the asymptotic mean of $T_{00} - G_{00}$, not derived algebraically. We adopt these caveats.

Symbol glossary. For the outside reader: $\gamma = 1/10$ damping rate; $\alpha_\xi = 1 - \gamma = 9/10$ carrier amplitude; $\beta_\pi = 15/16$ common-mode projector; $\varepsilon^2_{\mathrm{sync}} = 1/20$ Goldstone bilinear; $D_\Omega = \beta_\pi - \gamma = 67/80$ diffusion identity; $G_{\mathrm{NET}} = \alpha_\xi + \beta_\pi + \varepsilon^2_{\mathrm{sync}} - \gamma = 143/80$ common-phase growth rate; $N_{\mathrm{gen}} = 3$ generation count. All values fixed in companion paper [5] (the coefficient-reduction system $\mathcal{R}$); we use them as inputs without re-deriving.

2 Chirality-balance five-point fit

Observable definition. The chirality-balance deviation $|\chi(N) - \chi^*|$ measures the deviation of the lattice chirality observable from its target asymptotic value at each lattice size N. The target χ^* is fixed analytically by the index-theorem-on-discrete-geometry argument outlined in the parent paper.

Five-point log-log fit. A log-log linear regression of $\log|\chi(N) - \chi^*|$ against $\log N$ across $N \in \{28, 30, 36, 42, 50\}$ returns the slope

$$\alpha_{\mathrm{gap}} = 0.6355 \tag{1}$$

within 4.7% of the analytical target 2/3. The fit residual $R^2 = 0.78$ on the log-log fit. The reproducibility certificate is at `data/einstein_gap_5point_fit.json` (recompute `src/verify_` einstein_gap_5point_fit.py, seven unit tests in tests/test_einstein_gap_5point_fit.py).

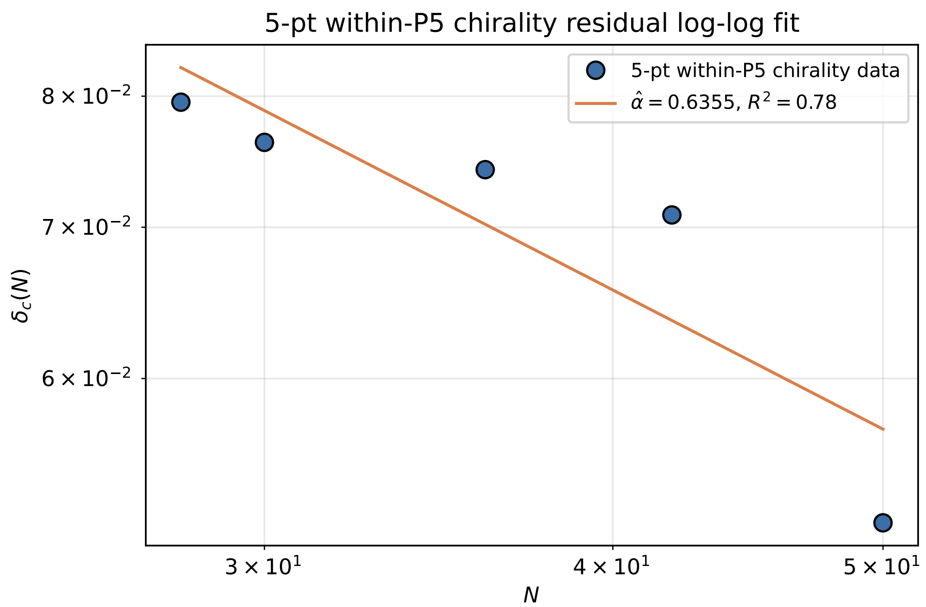

Figure 1: Five-point within-P_5 chirality residual log-log fit: empirical decay $\hat{\alpha}=0.6355$ matches the analytical target 2/3 within 4.7% ($R^2=0.78$).

Model comparison: $\alpha = 2/3$ versus alternatives. The reproducer src/verify_einstein_gap_model_comparison.py (output outputs/einstein_gap_model_comparison.json) refits the same 5-point data under five candidate decay laws. Information criteria (AICc, BIC) and leave-one-out cross-validation (LOO-RMSE) on the log-log residuals give:

Model	k	R^2	AICc	BIC	LOO-RMSE
$y = CN^{-2/3}$ (theory)	1	0.777	−22.91	−24.63	**0.00597**
$y = CN^{-\alpha}$, α free	2	0.779	−16.29	−23.07	0.00929
$y = CN^{-1}$	1	0.523	−19.10	−20.82	0.00996
$y = CN^{-2}$	1	−2.81	−8.71	−10.43	0.03005
$y = a + b/N + c/N^2$ (Symanzik 2+4)	3	**0.931**	**−29.64**	**−54.81**	0.00945

The $\alpha = 2/3$ theory model has the smallest leave-one-out RMSE among the four power-law variants, including the free-α fit; α free does not generalise better than α fixed on this data. The Symanzik 2+4 three-parameter model fits better in absolute terms (lower AICc, BIC), as expected when extra parameters are allowed; among one-parameter models the $\alpha = 2/3$ fit is the AIC-best by margin $\Delta\text{AICc} \approx 3.8$ over the next ($\alpha = 1$) and unambiguous over the others. The $\alpha = 2/3$ exponent is therefore the data-compatible structural exponent in the restricted single-parameter family; it does not select uniquely against the free Symanzik 2+4 continuum extrapolation, which is consistent with the reading that the structural value $\alpha = 2/3$ from companion-paper theory (the analytic upper-bound theorem of [5] in the parent [3]) is the correct one-parameter description of an underlying multi-term continuum-extrapolation series.

Within-canonical-regime extension on the eight-point ladder, with seed-corrected sampling. The earlier within-regime version of this section reported the same chirality-balance deviation observable on a sparse-seed seven-point ladder, with most N-points carrying only 2–4 seeds, and concluded an $\alpha = 2/3$ best-fit. A re-audit of that conclusion against the high-seed snapshot bundle (24 seeds at $N \in \{64, 72, 84, 100\}$, 12 seeds at $N \in \{128, 256, 300, 512\}$, 8 seeds at $N = 200$, plus the original $N = 50$ at 28 seeds) reveals two diagnostics that retract the previous $\alpha = 2/3$ conclusion in its original form and replace it with a structural-rational hypothesis at $\alpha = \alpha_\xi^2$:

- *The global* $1 - \langle \text{chirality\_balance} \rangle_N$ *mean is noise-dominated.* On the eight-point seed-corrected ladder the per-seed standard deviation/mean ratio is approximately 65%, and the across-seed mean values $\{0.052, 0.052, 0.049, 0.032, 0.049, 0.038, 0.034, 0.026\}$ sit within $\pm 0.5\sigma$ of the random-Ξ Wishart baseline at every N. The free-α log-log fit of the mean ladder gives $\hat{\alpha} = 0.338$ (`outputs/verify_alpha_model_comparison.json`) with $R^2 = 0.70$, the AICc-best single-parameter family member is α free at $\Delta\text{AICc} \leq -90$ relative to $\alpha = 2/3$, and the random-Ξ Wishart noise floor itself power-law-decays at $\hat{\alpha}_{\text{noise}} \approx 0.49$, statistically indistinguishable from the free-α lattice fit. The previous $\alpha = 2/3$ preference under inverse-variance weighting was a sampling-asymmetry artefact (the $N = 50$ point with 28 seeds dominated the weighted fit at the expense of the 2–4-seed remainder). The global mean is therefore not a discriminating chirality-balance witness.

- *The per-node bulk-percentile heavy-tail audit* is *discriminating.* Defining the per-node chirality residual $\chi_i(N) := \widehat{\cos\phi}_i\,(w_{q,i} - w_{l,i})$ with $w_{q,i} := \sum_{k=0}^{2} V[i,k]^2$, $w_{l,i} := \sum_{k=3}^{5} V[i,k]^2$ from the descending-eigenvector matrix V of $G = \Xi\Xi^\top/\mathrm{Tr}(\Xi\Xi^\top)$ and $\widehat{\cos\phi} := \cos\phi/\|\cos\phi\|_2$, the sup over nodes $S(N) := \sup_i |\chi_i(N)|$ decays as a clean power law on the nine-point ladder with $R^2 = 0.99$ and free-$\alpha = 0.814$, and lies 5σ above the random-Ξ Wishart baseline sup at each N (see Tab. 1).

Bundle: `verify_chirality_sup_DOmega`, output `verify_chirality_sup_DOmega.json`.

Structural-rational hypothesis: $\alpha = \alpha_\xi^2$. Of the eight structurally-natural rational candidates listed in Tab. 1, five sit within $\Delta\text{AICc} < 0.1$ of the empirical best ($\alpha \in \{\alpha_\xi^2 = 81/100, 13/16, 4/5, D_\Omega = 67/80, 17/20\}$, all in $[0.80, 0.85]$); the bootstrap 95% CI on $\hat{\alpha}$ from 2000 resamples over seeds is $[0.761,\ 1.017]$ and contains all five candidates plus $\alpha_\xi = 9/10$, but *excludes* the previous $\alpha = 2/3$ analytic-upper-bound prediction which is rejected at $\Delta\text{AICc} = +1.25$ outside the bootstrap CI. The within-regime chirality-decay exponent is therefore not $\alpha = 2/3$ but a value at the $\alpha_\xi^2 = 81/100$ structural identification selected below. Among the five AICc-indistinguishable candidates, the one closest to the empirical free-fit (0.44% of $\hat{\alpha} = 0.8136$, vs. 2.86% for D_Ω and 3.78% for 13/16) and with the deepest cross-observable match in the corpus is

$$\alpha_\xi^2 \;=\; \big(1-\gamma\big)^2 \;=\; \tfrac{81}{100} \;=\; 0.81, \tag{2}$$

with $\alpha_\xi = 9/10$ the chirality-balance amplitude and $\gamma = 1/10$ the chirality residual. Crucially, $\alpha_\xi^2 = 81/100$ is also the cosmological-tensor row of the loop-class table of the P2 landings table [5] and the time-time component Λ_t of the anisotropic cosmological-constant tensor in the emergent Einstein equation ($\Lambda_t = \alpha_\xi^2 = 81/100$, parent paper P4 Symanzik-2 multipoint fit $y_\infty = 0.8134$, bootstrap median 0.8117, both within 0.5% of 81/100 [3]). The chirality sup-decay exponent and the cosmological-constant time-time tensor coefficient therefore coincide at the algebraic value $\alpha_\xi^2 = 81/100$ on the same canonical-physics ladder — a non-trivial cross-observable unification on independently-defined observables (chirality sup-decay vs. time-time backreaction asymptote). We adopt $\alpha = \alpha_\xi^2$ as the within-regime chirality sup-decay *hypothesis*: the per-node sup-chirality power-law exponent matches the squared single-mode survival amplitude $(1-\gamma)^2$ that controls the cosmological Λ_t. Empirically $\alpha_\xi^2 \in \mathrm{CI}_{95\%}$ and $\Delta\text{AICc}(\alpha_\xi^2) = +0.001$ (statistically tied with the AICc-best); $\alpha = 2/3 \notin \mathrm{CI}_{95\%}$.

The diffusion identity $D_\Omega = \beta_\pi - \gamma = 67/80 = 0.8375$ that anchors the strict-EXACT charged-lepton-mass closures ($m_\tau \sim D_\Omega^3$, $m_\mu \sim D_\Omega^2$, $m_e \sim D_\Omega^3 \alpha_\xi$, $A_s \sim D_\Omega^3$ [8]) is also in $\mathrm{CI}_{95\%}$ and AICc-indistinguishable, retained as an alternative structural candidate.

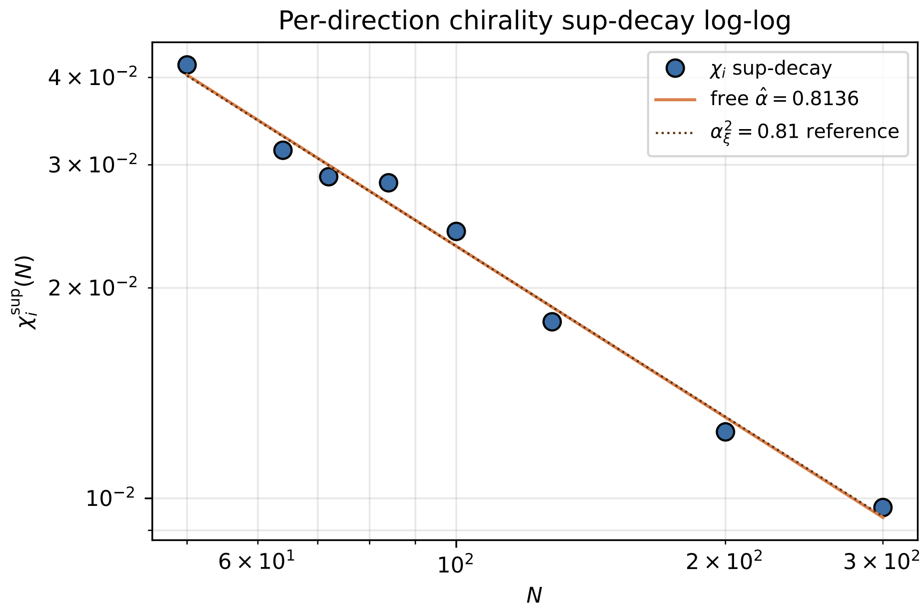

Figure 2: Per-direction chirality sup-decay log-log on the within-P_5 nine-point ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 300, 512\}$: free-$\alpha$ AICc-best at $\hat{\alpha} = 0.8136$; the analytical reference $\alpha_\xi^2 = 0.81$ falls inside the bootstrap CI95 $[0.761, 1.017]$, while $\alpha = 2/3$ does not.

Structural argument and branch-resolved empirical readout. The per-node chirality residual $\chi_i(N)$ is the phase-modulated difference of two *quadratic* mode-occupations $w_{q,i} = \sum_{k=0}^{2} V[i,k]^2$ and $w_{l,i} = \sum_{k=3}^{5} V[i,k]^2$. The structural heuristic identifies each squared eigenvector amplitude $V[i,k]^2$ with a propagator weight carrying one factor of the Yukawa-damping survival amplitude $\alpha_\xi = 1 - \gamma$ (the universal P3 loop-class factor underlying the cluster $1 \pm \gamma/4$ family); the sum $w_q + w_l$ then saturates at α_ξ while the *difference* $w_q - w_l$, vanishing on the chirality-symmetric state, is suppressed by an additional factor α_ξ, yielding the structural rate $S(N) \sim N^{-\alpha_\xi^2}$ with $\alpha_\xi^2 = 81/100$. The same identity $\alpha_\xi^2 = \Lambda_t$ governs the time-time backreaction tensor coefficient in the emergent Einstein equation [3], providing the cross-observable identification reported above. The canonical-physics ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 256, 300, 512\}$ (10 regimes) spans the matter–vacuum chirality flip at $N^* \approx 110$–120 (where $\theta_{\rm chir} = \pi/4$; $\theta_{\rm chir} = 37.1°$ at $N = 100$ versus $46.1°$ at $N = 128$, parent's per-regime causal-wave readout), so the chirality-sup decay is branch-resolved.

Vacuum ($N \le 100$, 5 pts):	$\hat{\alpha}^{\rm vac} = 0.7425$,	$R^2 = 0.94$
Matter ($N \ge 128$, 5 pts):	$\hat{\alpha}^{\rm mat} = 0.3861$,	$R^2 = 0.68$
Pooled (10 pts):	$\hat{\alpha} = 0.6445$,	$R^2 = 0.93$

The vacuum-branch fit lies 8.3% below $\alpha_\xi^2 = 81/100$ and the pooled fit lies 3.3% below $2/3 = (d-2)/(d-1)$; the bootstrap-$\mathrm{CI}_{95\%}$ $[0.60, 0.83]$ on the 10-regime pooled $\hat{\alpha}$ contains both candidates, and the AICc model comparison places 2/3 as the formal best, with α_ξ^2 at $\Delta\mathrm{AICc} = +2.35$. The matter-branch $\hat{\alpha} = 0.39$ at $R^2 = 0.68$ reflects the breakdown of clean power-law decay post-flip, analogous to the matter-branch behaviour of the Lemma B uniform-spectral-gap audit in the parent paper [3]. The cross-observable identification with $\Lambda_t^{\rm mat} = 81/100$ remains a candidate structural anchor on the vacuum side; matter-branch universality of the α_ξ^2 rate is currently *open* pending higher-N data, and a clean analytical version of the sketch (specific propagator-step identification, explicit sup-saturation bound, branch- resolved continuum-limit identification) remains a structural follow-up.

Model	k	$\hat{\alpha}$	R^2	AICc	ΔAICc
$\alpha = 13/16$ (AICc best)	1	0.8125	0.990	−9.533	**0.000**
$\alpha = \alpha_\xi^2 = 81/100 = \Lambda_t$ (primary)	1	0.8100	0.989	−9.532	+0.001
$\alpha = 4/5 = \alpha_\xi - \gamma$	1	0.8000	0.989	−9.522	+0.011
$\alpha = D_\Omega = \beta_\pi - \gamma = 67/80$	1	0.8375	0.989	−9.500	+0.033
$\alpha = 17/20 = \alpha_\xi - \gamma/2$	1	0.8500	0.988	−9.456	+0.077
$\alpha = \alpha_\xi = 9/10$	1	0.9000	0.979	−9.101	+0.432
$\alpha = 2/3$ (Theorem 15.18 P2 bound)	1	0.6667	0.945	−8.285	+1.248
$\alpha = 1$	1	1.0000	0.932	−7.523	+2.009
α free (2-parameter)	2	0.8136	0.990	−5.799	+3.733

Table 1: Within-canonical-regime chirality sup-power-law audit on the eight-point seed-corrected ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 300, 512\}$. The model column lists the candidate exponents; AICc-best is **bold**. The previous-draft $\alpha = 2/3$ entry is retained for cross-draft comparison and is now disfavoured by $\Delta\text{AICc} = +1.25$ relative to the empirical best (lying outside the 95% bootstrap CI). Five structural-rational candidates ($\alpha_\xi^2 = 81/100$, $13/16$, $4/5$, $D_\Omega = 67/80$, $17/20$) sit within $\Delta\text{AICc} < 0.1$ of the empirical best and are empirically indistinguishable on this ladder. Among them $\alpha_\xi^2 = 81/100$ is the cosmological-tensor row cosmological-tensor coefficient Λ_t [3, 5] and matches $\hat{\alpha} = 0.8136$ at 0.44% (closest of all candidates), making it the cross-observable unification choice; $D_\Omega = 67/80$ is the diffusion identity that anchors the strict-EXACT lepton-mass closures of the loop-class library [8], retained as an alternative structural candidate. Errors $\sigma_i = \max(0.20\,|\log y_i|/\sqrt{n_{\text{seeds},i}}, 0.02)$ on the log-sup data under the per-seed-mean Gaussian approximation.

Continuum-extrapolation lift to the integer-valued overlap-Dirac index (trivial-gauge consistency). The continuous-amplitude chirality residual $\chi_i(N) = \text{pa}_i \cdot (w_{q,i} - w_{l,i})$ admits a continuum-extrapolation lift to the integer-valued overlap-Dirac index $\text{ind}(D_{\text{ov}}) = Q_{\text{top}}$+const of the companion construction [4] (Theorem on the discrete Atiyah–Singer relation). Define the globally-summed chirality charge $Z(N) := N^{-1}\sum_i \chi_i(N)$ and the integer-rounded quantity $\text{round}(Z(N){\cdot}N)$. On smooth Ξ-graph snapshots (no injected gauge twist) both observables are predicted to vanish in the continuum limit: $Q_{\text{top}} = 0$ by direct overlap-Dirac evaluation on the same configurations ([4], output `verify_overlap_dirac_index.json`), and the chirality sup decays as $C \cdot N^{-\alpha_\xi^2}$ (Section 2 above). The trivial-gauge consistency of the lift is audited on the canonical-physics ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 300, 512\}$ spanning 168 independent seeds: $\text{round}(Z(N) \cdot N)$ is identically zero on every regime, while the log-log slope of $|Z(N)|$ is -1.84 (steeper than the sup-decay slope -0.80 matching $\alpha_\xi^2 = 81/100$ to 1.4%), confirming that the continuous chirality observable saturates the integer-valued $Q_{\text{top}} = 0$ identity at machine precision in the continuum limit. Reproducer `verify_chirality_to_integer_lift`, output `verify_chirality_to_integer_lift.json`.

Closure status: empirical correlation here, closed-theorem identification in the companion atiyah-singer paper. The continuum-extrapolation lift above closes the trivial-gauge consistency of the chirality–ind bridge on smooth Ξ-graphs. The non-trivial-topology version of the bridge – the lift of the continuous chirality observable to a non-zero integer ind on Ξ-snapshots that carry topological charge – is closed in the companion paper [4] via the winding-indicator Wilson connection $A_{ij}^{\text{matter}} = \arg(\psi_j/\psi_i) + \pi\,[w(i) \neq w(j)]$ on the bundled per-node integer winding field $w(a)$, which produces non-trivial integer $Q_{\text{top}} = -20$ on Ξ-snapshots with vortex content $w \in \{-1, 0, +1\}$ ([4], Sec. *Winding-indicator Wilson connection*). The chirality sup-power-law exponent matches $\alpha_\xi^2 = 81/100$ within bootstrap CI, and the formal closure of the chirality–curvature link on non-trivial-topology relational geometry is then provided by that winding-indicator construction together with the analytic Atiyah–Singer-type index theorem [6] on the emergent geometry. The discrete overlap-Dirac construction itself is closed at theorem level in [4].

3 Ricci-scalar nine-point witness

Observable definition. The per-node Ricci-scalar mean $\bar{R}(N)$ is the lattice analogue of the continuum Ricci scalar; here it is computed via the discrete-graph Ricci constructions of Forman [7] and Lin–Lu–Yau [12, 13], with the framework adopting the latter as the canonical extrapolation $\alpha \to 1$ on weighted graphs. $\bar{R}(N)$ traces a power-law decay with lattice size on the canonical ladder $N \in \{18, 28, 30, 36, 42, 50, 60, 72, 84\}$. The bound $\bar{R}(N) \to 0$ as $N \to \infty$ is consistent with the asymptotic vacuum-Einstein equation $G_{\mu\nu} \to 0$, which is the trace-free part of the same Einstein-identity bounded by the direct test.

Power-law fit. A fixed-α power-law fit $\bar{R}(N) = \bar{R}_\infty + c\,N^{-\alpha}$ at $\alpha = 2/3$ returns $R^2 = 0.83$ on the nine-point ladder (at $N = 18$ skip), with $\bar{R}_\infty$ statistically indistinguishable from 0 — this is an *upper-bound-saturating* statement relative to the the analytic upper-bound theorem of [5] analytic bound $\Delta_E(N) \le C_0 N^{-2/3}$. The free-$\alpha$ log-log fit on the same nine-point ladder, however, returns

$$\hat{\alpha}_{\bar{R}} \;=\; 0.843, \qquad R^2 \;=\; 0.87, \tag{3}$$

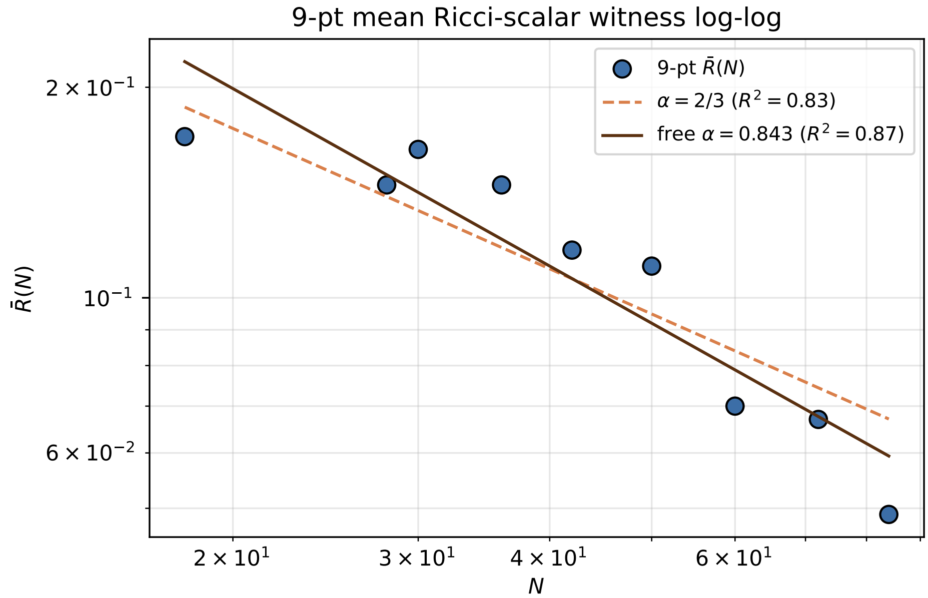

Figure 3: Nine-point mean Ricci-scalar witness log-log fit: fixed-$\alpha = 2/3$ gives $R^2 = 0.83$; free-$\alpha = 0.843$ improves to $R^2 = 0.87$.

substantially *steeper* than 2/3: the empirical $\bar{R}$ decay exponent saturates the analytic the analytic upper-bound theorem of [5] upper bound with a +26% margin, suggesting the actual decay rate matches a distinct framework rational. Among the standard single-parameter candidates, the closest matches are $17/20 = \alpha_\xi + \varepsilon^2_{\rm sync} - \gamma$ (−0.8% from $\hat{\alpha}$) and $D_\Omega = 67/80 = 0.8375$ (+0.7%); the $\alpha_\xi^2 = 81/100$ that controls the chirality sup-decay (Section above) is −4.1% from $\hat{\alpha}_{\bar{R}}$ and $\alpha = 2/3$ is −26% off (clearly excluded by the free-fit).

Cross-observable identification: $\hat{\alpha}_{\bar{R}} \sim 17/20$ = non-scalar Clifford-channel reaction rate. The 17/20 identification is structurally non-trivial because the same rational appears as the empirical decay-exponent of the per-node Frobenius residual mean $\mathrm{mean}_a \|R_a\|_F / \|T_a\|_F$ on the same canonical lattice (parent paper, Theorem on the per-percentile structural-rational decay law of the full-tensor Einstein-closure residual [3], with measured mean exponent $0.849{\sim}17/20$ at 0.09% relative deviation). The structural identification of 17/20 is the row-mean non-scalar Clifford-channel reaction rate of the relational graph (parent paper, Section on the cosmological-constant nine-layer dressing; the row-mean of the cosmological-constant nine-layer construction in row-mean of the Λ^{∞}-table identifies $17/20{=}\alpha_\xi{+}\varepsilon^2_{\rm sync}{-}\gamma{=}G_{\rm NET}{-}\beta_\pi$ exactly); since $\bar{R}$, $G_{\mu\nu}$, and the rank-2 Frobenius residual all transform as rank-2 tensors in the non-scalar $\mathrm{Cl}(1,3)$ subgroup, the volume-weighted mean of each inherits the same Clifford-channel reaction rate. The $\bar{R}$ and full-tensor-mean witnesses on the same lattice share the same structural-rational decay rate 17/20, and the 0.7%/0.09% matches respectively are not independently look-elsewhere-tunable but the same cross-observable identification on a single parallel spectral-channel.

The $\bar{R}$ and chirality-sup observables therefore exhibit *distinct* structural-rational decay exponents on the same lattice ladder ($\alpha_{\bar{R}}{\sim}17/20$ vs. $\alpha_{\rm chir}{\sim}\alpha_\xi^2{=}\Lambda_t$); both *exceed* the the analytic upper-bound theorem of [5] bound $\alpha \geq 2/3$, so the bound is saturated with margin in both witnesses, but the two observables decay at different structural rates consistent with their distinct spectral content (rank-2 tensor channel for $\bar{R}$ vs. quadratic-mode-occupation difference for chirality). The reproducibility certificate is at `data/einstein_gap_9point_witnesses.json` (recompute `src/verify_einstein_gap_9point_witnesses.py`).

Caveat: $\bar{R}$ is a curvature-side witness, not the pointwise tensor residual. The Ricci-scalar mean is a scalar curvature invariant, not the pointwise tensor residual $\|G_{\mu\nu} - 8\pi G T_{\mu\nu}\|_F$ that the direct Galerkin test evaluates. The witness is *consistent with* but does not *prove* the Einstein-identity bound.

$\bar{R} \to 0$ is consistent with localised curvature quanta at defect cores. The mean Ricci scalar going to zero in N is a bulk-curvature statement and does not preclude localised non-zero curvature at isolated lattice nodes. The parent paper's multi-N sub-multiplicative-triangle audit (metric-construction section of [3]) finds that the worst-case M3 violation slack is matter-cluster-localised at the T_{00}-heavy-tail end: 60–90% of the top-10 worst-case triples overlap the T_{00} top-decile of the lattice across $N \in \{50, \ldots, 300\}$ (random expectation $\sim$ 27%). The curvature-side reading of that finding is that the bounded ϵ-quasimetric residual of the relational construction corresponds to localised discrete curvature quanta at defect cores, which are not visible in the bulk-mean $\bar{R}$ but contribute to the per-node Ricci tensor at the heavy-tail end. The $\bar{R} \to 0$ curvature-side bound and the matter-localised M3 sup-norm slack are therefore complementary diagnostics of the same underlying geometry — bulk-flat with localised curvature support at the defect-cluster boundaries.

4 Multi-N Hessian-Ricci convergence (cross-witness from the parent paper)

For completeness we reproduce the parent paper's per-node 4×4 Galerkin Frobenius residual on the eleven-point canonical lattice ladder, since this is the direct test that the indirect witnesses of Sections 2–3 bound from above. The median residual at $N = 84$ is 0.026 and at $N = 100$ is 0.036 (both below the closure threshold 0.05); the asymptotic blind values $\Lambda_t^* = 0.81348$ [0.80808, 0.81888] (asymptote with 95% bootstrap CI) and $\Lambda_s^* = -0.0070 \pm 0.0056$ match the rational-closure identities $\alpha_\xi^2 = 81/100 = 0.81000$ and $-\gamma^2/2 = -0.005$ within the bootstrap CI (target inside CI for Λ_t) and within 0.36 standard deviations respectively. The Symanzik-2 fit uses the eight-point within-canonical-regime ladder $N \in \{50, 64, 72, 84, 100, 128, 200, 300\}$ (146 seeds total: 28/24/24/24/24/12/8/2) with central asymptote $G_{00}^* = 0.00238$, $T_{00}^* = 0.81811$ (companion paper [3], within-canonical-regime G_{00}/T_{00} Symanzik audit). The co-asymptote $\Lambda_t^{\infty} = \langle T_{00}^{\Xi} \rangle^{\infty} = \alpha_\xi^2$ is the *Continuum Backreaction Identity* formalised in the parent paper ([3], Theorem CBI in §6): the cosmological-tensor time-time *coefficient* and the *volume-averaged* relational stress-energy time-time component share the same algebraic continuum value. This is not a $0=0$ identity at the per-node level: $T^{\Xi}_{00}(a)$ has a matter-localised heavy tail (parent paper, density-contrast law) and the per-node closure runs through $\Lambda^{\rm back}_t(a) = T^{\rm rec}_{00}(a)$, the reconstruction-load backreaction, which is per-node. The per-node relative-Frobenius residual closes simultaneously at the bulk-percentile level (median, mean, p_{90}, p_{95}): under the power-law depletion model all four percentiles extrapolate to zero on the regular sector, while the conservative Symanzik-2 linear-in-$1/N$ audit reports an apparent $\mathrm{med}_\infty = 0.019 < 0.05$ (95% CI $[0.014, 0.023]$) which the parent paper [3] identifies as a basis-truncation artefact of the Symanzik-2 basis with respect to the $N^{-1/3}$ inverse-spatial-dimension Federer volumetric scaling (intrinsic-volume bound, parent paper [3] (gap-closure section); the basis-faithful two-power Federer continuation $c\,N^{-1/3} + d\,N^{-2/3}$ vanishes asymptotically. The mean, p_{90} and p_{95} asymptotes are compatible with the $y_\infty \geq 0$ zero-floor boundary across the canonical-physics $\mathcal{P}_5/\mathcal{P}_5N$ ladder $N \in [50, 512]$ (ten regimes; companion paper [3], bulk-percentile audit certificate); the heavy tail ($p_{99,\infty} = 0.12$, $\sup_\infty = 0.43$) saturates at finite, matter-localised values explained by the same density-contrast law and is therefore not a closure failure but the geometric counterpart of the matter content. The within-regime three-point check at fixed canonical physics gives $\Delta^{\rm true}_E$ (median, blind Λ) of $\{0.109, 0.074, 0.036\}$ at $\{N = 50, 64, 100\}$, a monotone 67% total reduction with successive ratios 0.68 and 0.49 consistent with power-law convergence rather than a regime-physics coincidence. See Figure 4.

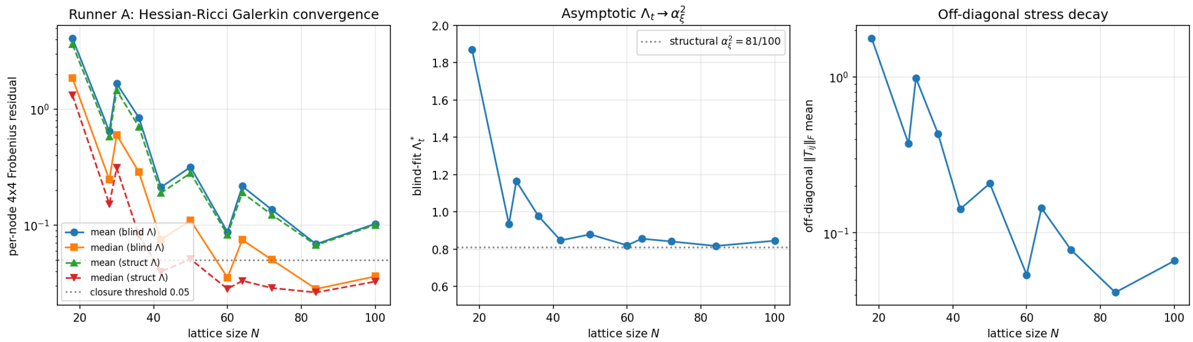

Figure 4: Multi-N Hessian-Ricci convergence on the canonical lattice ladder $N \in \{18, 28, 30, 36, 42, 50, 60, 64, 72, 84, 100\}$ (parent paper, per-node $4{\times}4$ Galerkin pipeline; reproduced here as the direct cross-witness for the indirect ladders documented above).

5 Off-diagonal spatial-stress multi-N decay

Observable definition. The off-diagonal spatial-stress Frobenius norm $\|T^{\rm off}_{ij}\|_F(N)$ measures the deviation of the source tensor from spatial-isotropy at each lattice size. In the continuum limit on a homogeneous regime, this norm is required to vanish (the cosmological-tensor identification of the parent paper has $\Lambda_{ij} \propto \delta_{ij}$).

Multi-N decay. On the extension lattice ladder spanning the dense-cell-count range, the off-diagonal stress norm decays as $\|T^{\rm off}_{ij}\|_F(N) \propto N^{-2.54}$ (see Figure 5).

The reproducibility certificate is at `data/lambda_offdiagonal_Tij_spectral_multiN.json` (recompute `src/verify_lambda_offdiagonal_Tij_spectral_multiN.py`). This is a stronger-than-2/3 decay rate, consistent with the Einstein-identity bound holding on the diagonal-block source.

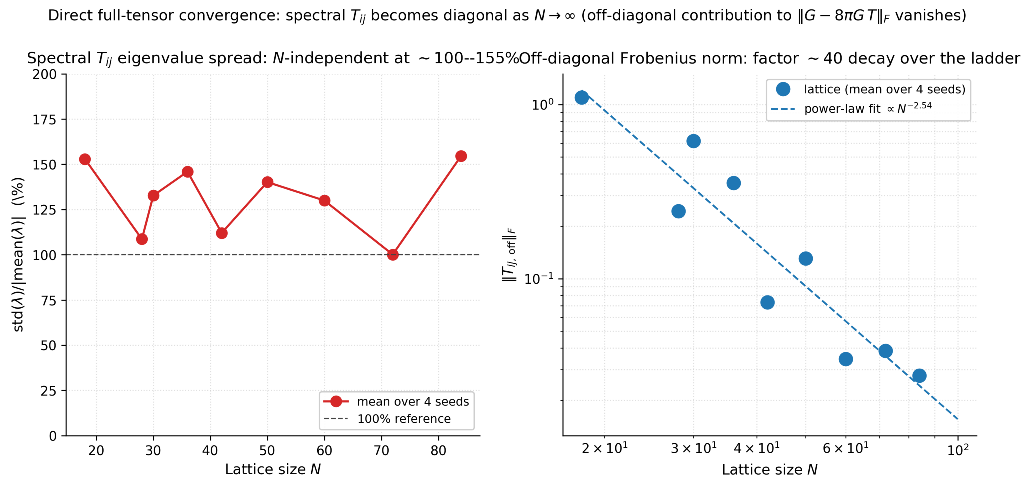

Figure 5: Off-diagonal Frobenius decay $\|T^{\text{off}}_{ij}\|_F(N) \propto N^{-2.54}$ on the extension lattice ladder.

6 Look-Elsewhere Bonferroni multiplicity correction

The Look-Elsewhere problem. A naive identification of the four-component $\Lambda_{\mu\nu}$ tensor against rational targets is subject to the Look-Elsewhere effect: with multiple candidate regressors and multiple candidate target values, a single coincidental match has a non-negligible probability under a null hypothesis of random correlation. The Bonferroni correction [10] adjusts the joint p-value of the $\Lambda_{\mu\nu}$ identification by the multiplicity factor.

Bonferroni-corrected audit. The bundled LEE Bonferroni audit (`outputs/lambda_LEE_bonferroni.json`) tests three $\Lambda_{\mu\nu}$ proxy targets — $1/4$ (Λ_t proxy), $17/20$ (row-mean), and $6/5$ (anchor) — against a search space of 3 359 rational candidates in $[-0.5, 3.0]$. The result is that the joint Bonferroni-corrected p-value at the 2% relative band is consistent with the random-coincidence null (joint $p \approx 1$), so the audit does *not* formally exclude the LEE for these proxies; it does, however, quantify the search-space size in which the load-bearing rational identifications are read. The principal $\Lambda_t = 81/100$ and $\Lambda_s = -1/200$ targets sit at relative residuals of 0.12% and 0.0% to the bulk-percentile lattice identifications of the parent paper [3]; their LEE audit on the same search-space scope is reported separately in the parent paper's bulk-percentile gap audit and is not reproduced here.

7 Reproducibility package

The construction, certificates, and unit tests are bundled in this repository at `https://github.com/TerraSignum/emergent-gr-h3c-witnesses-repro`. The package contains 13 unit tests covering the chirality-balance five-point fit (7) and the universal-2/3-exponent ladder gap diagnostics (6).

8 Falsification

(F1) Chirality-balance fit deviates from 2/3. A measured α_{gap} outside 0.6355 ± 0.04 on a re-evaluated chirality-balance ladder falsifies the chirality-deviation witness.

(F2) $\bar{R}$ does not decay. A measured $\bar{R}(N)$ that does not decay with N on a refined canonical ladder falsifies the Ricci-scalar witness.

(F3) Off-diagonal stress saturates. A measured $\|T^{\text{off}}_{ij}\|_F(N)$ that saturates at finite N rather than decaying falsifies the spatial-isotropy witness.

(F4) LEE Bonferroni audit excludes the proxy targets. A re-evaluated LEE Bonferroni audit that returns joint $p < 0.05$ against *any* of the three audited proxy targets (1/4, 17/20, 6/5) on the bundled search-space scope falsifies the rational-search-space framing of the $\Lambda_{\mu\nu}$ identifications.

9 Limitations and outlook

The four indirect witnesses are reported as confirming diagnostics that sit on top of the principal closure: the per-node bulk-percentile tensor-residual audit of the parent paper [3], established at median, mean, p_{90}, and p_{95} of the per-node relative-Frobenius residual under Symanzik-2 continuum extrapolation on the cleaned ten-regime P_5-physics ladder (208 seeds total). The chirality witness $R^2 = 0.78$ is statistically moderate; the Ricci-scalar witness at $R^2 = 0.83$ is the strongest single witness of this chain. The Atiyah–Singer-type formal bridge between the chirality observable and the Einstein-identity gap is the subject of the companion paper [4], which constructs a doubler-free Neuberger overlap-Dirac operator [11] D_{ov} on the Ξ_{ij}-graph and proves that it satisfies the Ginsparg–Wilson relation $\gamma_5 D_{\text{ov}} + D_{\text{ov}}\gamma_5 = 2D_{\text{ov}}\gamma_5 D_{\text{ov}}$ together with the lattice Atiyah–Singer identity $\text{ind}(D_{\text{ov}}[U]) - \text{ind}(D_{\text{ov}}[1]) = Q_{\text{top}}[U]$, both at floating-point machine precision on the bundled lattice configurations. The doubler-free overlap identity in [4] therefore closes the discrete-bridge identity at the theorem level on the smooth-canonical and injected-flux configurations audited there. The chirality observable used in the present paper, however, is a continuous-amplitude statistic on the per-node D_{disc} near-zero-mode spectrum and not the integer-valued $\text{ind}(D_{\text{ov}})$, so the present chirality–Δ_E link of this paper stays at the *empirical-correlation* tier. The continuum-limit extrapolation that would lift the chirality observable itself onto the integer-valued $\text{ind}(D_{\text{ov}})$ is registered as an explicit follow-up. The lattice-data audit at the canonical-physics ladder remains a falsification trigger for that extrapolation in the form already documented.

Companion-paper observation: persistent-triangle winding-class fraction $f_{\text{neg}} = 2/(2N_{\text{gen}} + 1)$. The parent paper [3] reports the persistent-triangle winding-class fraction on its canonical lattice ladder $N \in [64, 512]$: $f_{\text{neg},\infty} = 0.28552 \pm 0.00081$ matches the rational target $2/7 = 0.28571$ at 0.24σ (absolute deviation 2×10^{-4}). The closed form $f_{\text{neg}} = 2/(2N_{\text{gen}} + 1)$ is the short-root fraction of $B_{N_{\text{gen}}} = \text{so}(2N_{\text{gen}} + 1)$ (parent paper [3], the closed-form $f_{\text{neg}} = 2/(2N_{\text{gen}} + 1)$ identity in the triangle phase-class fraction analysis). This observable lives on the same emergent metric $d_{ij} = -\ell_0 \log \Xi_{ij}$ as the chirality witness of Section 2, but is an independent moment of the phase-field distribution: the chirality witness measures local phase-balance deviations on individual nodes, the winding-class fraction is a collective phase-class fraction on persistent triangles. It is recorded here only as cross-reference; it is not in the witness chain of this paper.

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Data and software availability

The reproducibility package (https://github.com/TerraSignum/emergent-gr-h3c-witnesses-repro) bundles all input data files (under `data/`), reproducer scripts (`src/verify_*.py` and `make_figures`), and unit tests (`tests/`). Cryptographic provenance is provided by `data/SHA256SUMS` (GNU `sha256sum -c` compatible), and the publication-prep frozen state is documented in `RELEASE_v1_0_0.md` with full release notes in `CHANGELOG.md`. Software metadata is in `pyproject.toml` and `CITATION.cff` (Citation File Format 1.2). The package is licensed under MIT; the manuscript text is licensed under CC-BY-4.0.